

OCR Further Mathematics
Pure Core - Polar Coordinates
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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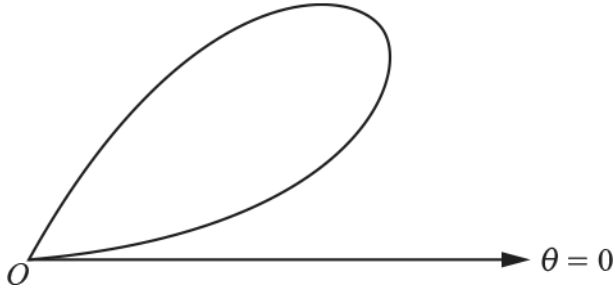
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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a)

The diagram below shows the curve with polar equation $r = \sin 3\theta$ for $0 \leq \theta \leq \frac{1}{3}\pi$.



Find the values of θ at the pole.

[1]

(b) Find the polar coordinates of the point on the curve where r takes its maximum value.

[2]

(c) In this question you must show detailed reasoning.

Find the exact area enclosed by the curve.

[4]

(d) Given that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$, find a cartesian equation for the curve.

[2]

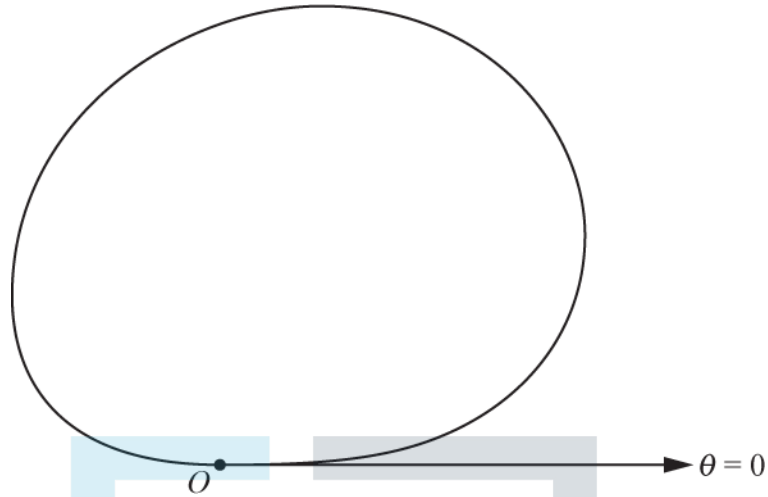
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2(a)

The diagram below shows the curve $r = \sqrt{\sin\theta}e^{\frac{1}{3}\cos\theta}$ for $0 \leq \theta \leq \pi$



Find the exact area enclosed by the curve.

[4]

(b)

Show that the greatest value of r on the curve is $\sqrt{\frac{\sqrt{3}}{2}}e^{\frac{1}{6}}$

[7]

- 3(a) The equation of a curve in polar coordinates is $r = \ln(1 + \sin \theta)$ for $\alpha \leq \theta \leq \beta$ where α and β are non-negative angles. The curve consists of a single closed loop through the pole.

By solving the equation $r = 0$, determine the smallest possible values of α and β .

[2]

- (b) Find the area enclosed by the curve, giving your answer to 4 significant figures.

[2]

- (c)

Hence, by considering the value of r at $\theta = \frac{\alpha + \beta}{2}$, show that the loop is not circular.

[2]

4(a) A curve has cartesian equation $x^3 + y^3 = 2xy$.

C is the portion of the curve for which $x \geq 0$ and $y \geq 0$. The equation of C in polar form is given by $r = f(\theta)$ for $0 \leq \theta \leq \frac{1}{2}\pi$.

Find $f(\theta)$.

[2]



(b)

Find an expression for $f\left(\frac{1}{2}\pi - \theta\right)$, giving your answer in terms of $\sin \theta$ and $\cos \theta$.

[2]

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(c) Hence find the line of symmetry of C .

[1]

(d)

Find the value of r when $\theta = \frac{1}{4}\pi$.

[1]



(e) By finding values of θ when $r = 0$, show that C has a loop.

[2]

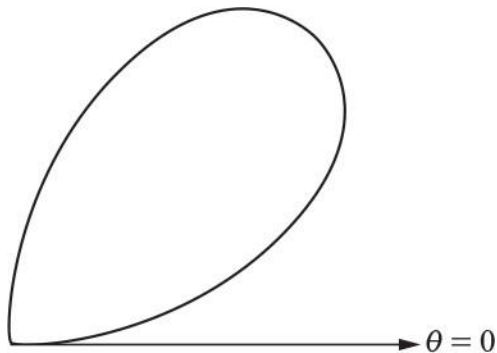
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5

The equation of the curve shown on the graph is, in polar coordinates, $r = 3 \sin 2\theta$ for $0 \leq \theta \leq \frac{1}{2}\pi$.



(a) The greatest value of r on the curve occurs at the point P .

(i) Show that $\theta = \frac{1}{4}\pi$ at the point P .

[2]

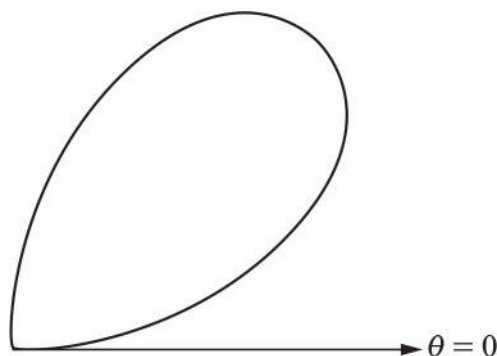
(ii) Find the value of r at the point P .

[1]

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(iii) Mark the point P on the copy of the graph below.

[1]



(b) In this question you must show detailed reasoning.

Find the exact area of the region enclosed by the curve.

[5]



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6 In this question you must show detailed reasoning.

S is the 2-D transformation which is a stretch of scale factor 3 parallel to the x -axis. A is the matrix which represents S .

(a) Write down A .

[1]

(b) By considering the transformation represented by A^{-1} , determine the matrix A^{-1} .

[2]

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Matrix B is given by $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. T is the transformation represented by B.

(c) Describe T

[1]

(d) Determine the matrix which represents the transformation S followed by T.

[2]

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(e) Demonstrate, by direct calculation, that $(BA)^{-1} = A^{-1}B^{-1}$.

[2]



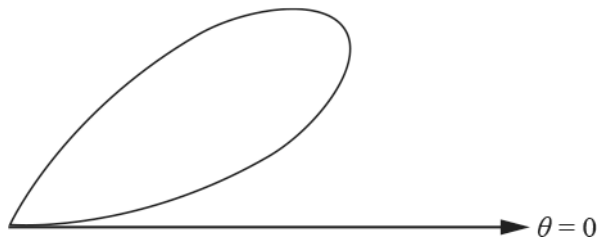
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7

The diagram below shows the curve $r = 4 \sin 3\theta$ for $0 \leq \theta \leq \frac{1}{3}\pi$.



(a) On the diagram above, shade the region R for which

$$r \leq 4 \sin 3\theta \text{ and } 0 \leq \theta \leq \frac{1}{6}\pi.$$

[1]

In this question you must show detailed reasoning.

(b) Find the exact area of the region R .

[4]

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- 8 (a) (i) A curve has polar equation $r = 2 - \sec \theta$. Show that the cartesian equation of the curve can be written in the form

$$y^2 = \left(\frac{2x}{x+1} \right)^2 - x^2.$$

[4]

The figure shows a sketch of part of the curve with equation $y^2 = \left(\frac{2x}{x+1} \right)^2 - x^2$.



- (ii) Explain why the curve is symmetrical in the x -axis.

[1]

(iii) The line $x = a$ is an asymptote of the curve. State the value of a .

[1]

(b) The enclosed loop shown in the figure is rotated through 180° about the x -axis.

Find the exact volume of the solid formed.

[8]



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9 The polar equations of two curves are $r = \sin \theta$ and $r = \cos 2\theta$.

(i) On the same diagram sketch the parts of both curves that are in the positive quadrant.

[4]



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The curves meet at the origin and also at a point A.

(ii) Determine the polar coordinates of A.

[2]

(iii) Find the exact area between the two curves in the positive quadrant.

[8]



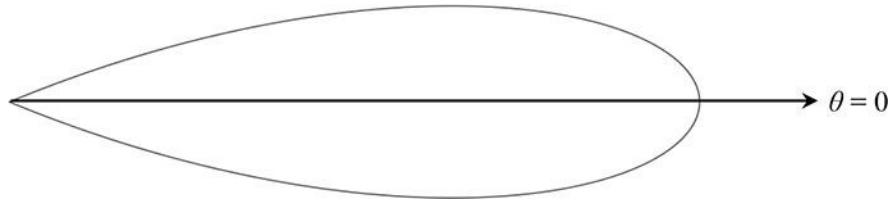
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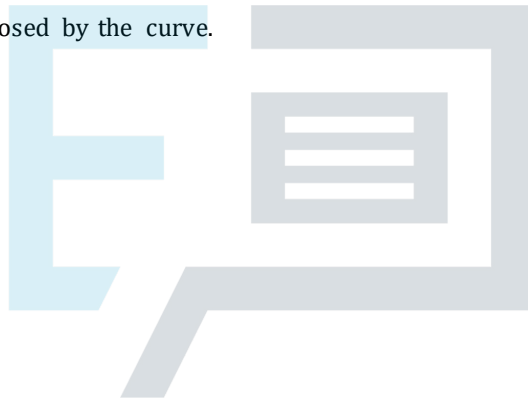
10 In this question you must show detailed reasoning.

The diagram below shows the curve $r = 2\cos 4\theta$ for $-\pi \leq \theta \leq \pi$ where k is a constant to be determined.



Calculate the exact area enclosed by the curve.

[6]



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11 A curve has equation $x^4 + y^4 = x^2 + y^2$, where x and y are not both zero.

(a) Show that the equation of the curve in polar coordinates is $r^2 = \frac{2}{2 - \sin^2 2\theta}$. [4]



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(b) Deduce that no point on the curve $x^4 + y^4 = x^2 + y^2$ is further than $\sqrt{2}$ from the origin. [2]

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The equation of a curve in polar coordinates is $r = \sin 5\theta$ for $0 \leq \theta \leq \frac{1}{5}\pi$.

- (i) Sketch the curve and write down the equations of the tangents at the pole.

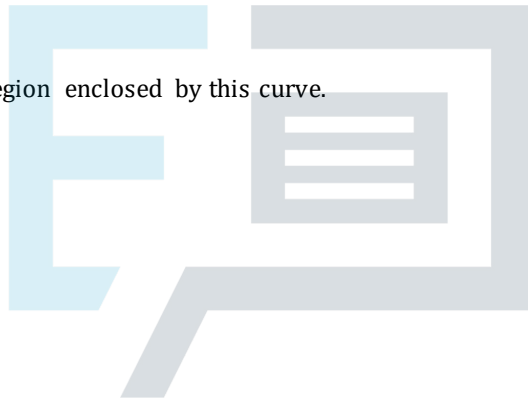
[4]

- (ii) The line of symmetry meets the curve at the pole and at one other point A. Find the equation of the line of symmetry and the cartesian coordinates of A.

[2]

- (iii) Find the area of the region enclosed by this curve.

[4]



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13 The equation of a curve in polar coordinates is $r = 2\sin 3\theta$ for $0 \leq \theta \leq \frac{1}{3}\pi$.

(i) Sketch the curve.

[2]

(ii) Find the area of the region enclosed by this curve.

[4]

(iii) By expressing $\sin 3\theta$ in terms of $\sin \theta$, show that a cartesian equation for the curve is

$$(x^2 + y^2)^2 = 6x^2y - 2y^3.$$

[5]

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14 A curve has polar equation $r = 5 \sin 2\theta$ for $0 \leq \theta \leq \frac{1}{2}\pi$.

(i) Sketch the curve, indicating the line of symmetry and stating the polar coordinates of the point P on the curve which is furthest away from the pole.

[4]

(ii) Calculate the area enclosed by the curve.

[3]

(iii) Find the cartesian equation of the tangent to the curve at P .

[3]

(iv) Show that a cartesian equation of the curve is $(x^2 + y^2)^3 = (10xy)^2$.

[3]

15 A curve has polar equation $r = a(1 + \cos\theta)$, where a is a positive constant and $0 \leq \theta < 2\pi$.

(i) Find the equation of the tangent at the pole.

[2]

(ii) Sketch the curve.

[2]

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© (iii) Find the area enclosed by the curve.

[6]

16 The equation of a curve is $x^2 + y^2 - x = \sqrt{x^2 + y^2}$.

(i) Find the polar equation of this curve in the form $r = f(\theta)$.

[3]

(ii) Sketch the curve.

[2]

(iii) The line $x + 2y = 2$ divides the region enclosed by the curve into two parts. Find the ratio of the two areas.

[6]



END OF QUESTION PAPER

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